

## 11. 表單

### 1. 表單樣式

form-control, form-label, form-text

```
<div class="demo">
  <form>
    <div class="mb-3">
      <label class="form-label" for="username">Username</label>
      <input type="text" id="username" class="form-control">
      <div class="form-text">不會記憶任何資訊!</div>
    </div>
  </form>
</div>
```

for, id 需要一致, 選取label也可同時選取input

Username

不會記憶任何資訊!

type="button" form表單無反應

type="submit" form表單內容送出並清空

```
<form>
  <div class="mb-3">
    <label class="form-label" for="username">Username</label>
    <input type="text" id="username" class="form-control">
    <div class="form-text">不會記憶任何資訊!</div>
  </div>
  <div class="mb-3">
    <label class="form-label" for="tel">電話</label>
    <input type="tel" id="tel" class="form-control">
    <div class="form-text">不會記憶任何資訊!</div>
  </div>
  <button type="button" class="btn btn-primary">確認</button>
  <button type="submit" class="btn btn-primary">確認</button>
</form>
```



利用 **align-items-center** 搭配 **form-text** 讓文字垂直置中

```
<div class="demo">
  <div class="row align-items-center">
    <div class="col-auto">
      <label class="form-label" for="username">Username</label>
    </div>
    <div class="col-auto">
      <input type="text" id="username" class="form-control">
    </div>
    <div class="col-auto">
      <div class="form-text">不會記憶任何資訊!</div>
    </div>
  </div>
</div>
```



**disabled**

```
<div class="demo">
  <form>
    <div class="row align-items-center">
      <div class="col-auto">
        <label class="form-label" for="username">Username</label>
      </div>
      <div class="col-auto">
        <input disabled type="text" id="username" class="form-control">
      </div>
      <div class="col-auto">
        <div class="form-text">不會記憶任何資訊!</div>
      </div>
    </div>
  </form>
</div>
```

Username  不會記憶任何資訊!

## 2. Form controls

```
<div class="demo">
  <input class="form-control form-control-lg mb-2" type="text" placeholder=".form-control-lg"
    aria-label=".form-control-lg example">
  <input class="form-control mb-2" type="text" placeholder="Default input" aria-label="default input
    example">
  <input class="form-control form-control-sm mb-2" type="text" placeholder=".form-control-sm"
    aria-label=".form-control-sm example">
</div>
```

.form-control-lg

Default input

.form-control-sm

## File input

```
<div class="demo">
  <div class="mb-3">
    <label for="formFile" class="form-label">Default file input example</label>
    <input class="form-control" type="file" id="formFile">
  </div>
  <div class="mb-3">
    <label for="formFileMultiple" class="form-label">Multiple files input example</label>
    <input class="form-control" type="file" id="formFileMultiple" multiple>
  </div>
  <div class="mb-3">
    <label for="formFileDisabled" class="form-label">Disabled file input example</label>
    <input class="form-control" type="file" id="formFileDisabled" disabled>
  </div>
  <div class="mb-3">
    <label for="formFileSm" class="form-label">Small file input example</label>
    <input class="form-control form-control-sm" id="formFileSm" type="file">
  </div>
  <div>
    <label for="formFileLg" class="form-label">Large file input example</label>
    <input class="form-control form-control-lg" id="formFileLg" type="file">
  </div>
</div>
```

Default file input example

選擇檔案 未選擇任何檔案

Multiple files input example

選擇檔案 未選擇任何檔案

Disabled file input example

選擇檔案 未選擇任何檔案

Small file input example

選擇檔案 未選擇任何檔案

Large file input example

選擇檔案 未選擇任何檔案

## Datalists

```
<div class="demo">
  <label for="exampleDataList" class="form-label">我的美食紀錄</label>
  <input class="form-control" list="datalistOptions" id="exampleDataList" placeholder="Type to search...">
  <datalist id="datalistOptions">
    <option value="牛肉麵">
    <option value="麥當勞">
    <option value="肯德基">
    <option value="必勝客">
    <option value="春水堂">
  </datalist>
</div>
```

我的美食紀錄

Type to search...

## 3. Select

```
<div class="demo">
  <select class="form-select" aria-label="Default select example">
    <option selected>Open this select menu</option>
    <option value="1">One</option>
    <option value="2">Two</option>
    <option value="3">Three</option>
  </select>
</div>
```

Open this select menu

Open this select menu

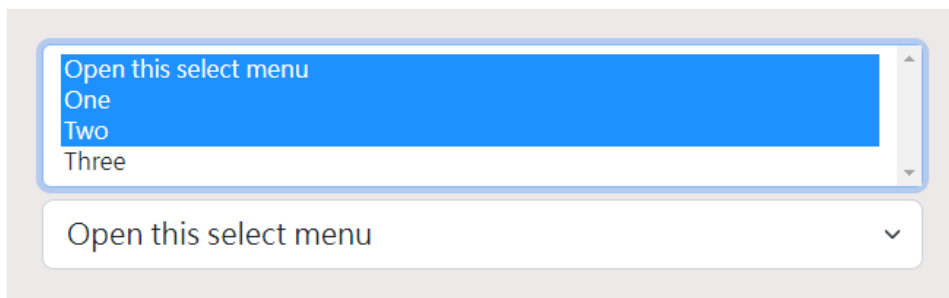
One

Two

Three

## Multiple, form-select-lg

```
<div class="demo">
  <select class="form-select mb-2" multiple aria-label="Default select example">
    <option selected>Open this select menu</option>
    <option value="1">One</option>
    <option value="2">Two</option>
    <option value="3">Three</option>
  </select>
  <select class="form-select form-select-lg" aria-label="Default select example">
    <option selected>Open this select menu</option>
    <option value="1">One</option>
    <option value="2">Two</option>
    <option value="3">Three</option>
  </select>
</div>
```



## Size 表達可以顯示幾筆資料

```
<select class="form-select mb-2" size="4" multiple aria-label="Default select example">
  <option selected>Open this select menu</option>
  <option value="1">One</option>
  <option value="2">Two</option>
  <option value="3">Three</option>
  <option value="4">Four</option>
</select>
```



## 4. Checks and radios

form-check, form-check-input, form-check-label,

```
<div class="demo">
  <div class="form-check">
    <input class="form-check-input" type="checkbox" value="" id="flexCheckDefault">
    <label class="form-check-label" for="flexCheckDefault">
      Default checkbox
    </label>
  </div>
  <div class="form-check">
    <input class="form-check-input" type="checkbox" value="" id="flexCheckChecked" checked="">
    <label class="form-check-label" for="flexCheckChecked">
      Checked checkbox
    </label>
  </div>
</div>
```

- ☐ Default checkbox
- ☒ Checked checkbox

## Radio: name 必須一致!

```
<div class="demo">
  <div class="form-check">
    <input class="form-check-input" type="radio" name="sameDefault" id="flexRadioDefault1">
    <label class="form-check-label" for="flexRadioDefault1">
      Default radio
    </label>
  </div>
  <div class="form-check">
    <input class="form-check-input" type="radio" name="sameDefault" id="flexRadioDefault2" checked="">
    <label class="form-check-label" for="flexRadioDefault2">
      Default checked radio
    </label>
  </div>
</div>
```

- ☐ Default radio
- ☒ Default checked radio

## Switches (form-switch, role="switch")

```
<div class="demo">
  <div class="form-check form-switch">
    <input class="form-check-input" type="checkbox" role="switch" id="flexSwitchCheckDefault">
    <label class="form-check-label" for="flexSwitchCheckDefault">Default switch checkbox input</label>
  </div>
  <div class="form-check form-switch">
    <input class="form-check-input" type="checkbox" role="switch" id="flexSwitchCheckChecked" checked="">
    <label class="form-check-label" for="flexSwitchCheckChecked">Checked switch checkbox input</label>
  </div>
  <div class="form-check form-switch">
    <input class="form-check-input" type="checkbox" role="switch" id="flexSwitchCheckDisabled" disabled="">
    <label class="form-check-label" for="flexSwitchCheckDisabled">Disabled switch checkbox input</label>
  </div>
  <div class="form-check form-switch">
    <input class="form-check-input" type="checkbox" role="switch" id="flexSwitchCheckCheckedDisabled" checked="" disabled="">
    <label class="form-check-label" for="flexSwitchCheckCheckedDisabled">Disabled checked switch checkbox input</label>
  </div>
</div>
```

- ☐ Default switch checkbox input
- ☒ Checked switch checkbox input
- ☐ Disabled switch checkbox input
- ☒ Disabled checked switch checkbox input

## Inline

```
<div class="demo">
  <div class="form-check form-check-inline">
    <input class="form-check-input" type="checkbox" id="inlineCheckbox1" value="option1">1</label>
  </div>
  <div class="form-check form-check-inline">
    <input class="form-check-input" type="checkbox" id="inlineCheckbox2" value="option2">2</label>
  </div>
  <div class="form-check form-check-inline">
    <input class="form-check-input" type="checkbox" id="inlineCheckbox3" value="option3" disabled="">3 (disabled)</label>
  </div>
</div>
```



☒ 1 ☒ 2 ☐ 3 (disabled)

```
<div class="demo">
  <div class="form-check form-check-inline">
    <input class="form-check-input" type="radio" name="inlineRadioOptions" id="inlineRadio1"
      value="option1">
    <label class="form-check-label" for="inlineRadio1">1</label>
  </div>
  <div class="form-check form-check-inline">
    <input class="form-check-input" type="radio" name="inlineRadioOptions" id="inlineRadio2"
      value="option2">
    <label class="form-check-label" for="inlineRadio2">2</label>
  </div>
  <div class="form-check form-check-inline">
    <input class="form-check-input" type="radio" name="inlineRadioOptions" id="inlineRadio3"
      value="option3"
      disabled>
    <label class="form-check-label" for="inlineRadio3">3 (disabled)</label>
  </div>
</div>
```

☐ 1 ☐ 2 ☐ 3 (disabled)

## Checkbox toggle buttons (保留狀態)

```
<div class="demo">
  <input type="checkbox" class="btn-check" id="btn-check" autocomplete="off">
  <label class="btn btn-primary" for="btn-check">Single toggle</label>
</div>
<div class="demo">
  <input type="checkbox" class="btn-check" id="btn-check-2" checked="" autocomplete="off">
  <label class="btn btn-primary" for="btn-check-2">Checked</label>
</div>
```

## 5. Range

```
<div class="demo">
  <label for="customRange3" class="form-label">Example range</label>
  <input type="range" class="form-range" min="0" max="100" step="1" id="customRange3">
</div>
<div class="demo">
  range的參數為: <span id="demo">0</span>
</div>
<script>
  const range = document.querySelector('#customRange3');
  range.addEventListener('change', (evt)=>{
    document.querySelector('#demo').innerHTML = evt.target.value;
  })
</script>
```

## 6. Input群組

### input-group, input-group-text

```
<div class="input-group mb-3">
  <span class="input-group-text" id="basic-addon1">@</span>
  <input type="text" class="form-control" placeholder="Username" aria-label="Username"
    aria-describedby="basic-addon1">
</div>
```

@ Username

```
<div class="input-group mb-3">
  <input type="text" class="form-control" placeholder="Recipient's username"
    aria-label="Recipient's username"
    aria-describedby="basic-addon2">
  <span class="input-group-text" id="basic-addon2">@example.com</span>
</div>
```

```
<label for="basic-url" class="form-label">Your vanity URL</label>
<div class="input-group mb-3">
  <span class="input-group-text" id="basic-addon3">https://example.com/users/</span>
  <input type="text" class="form-control" id="basic-url" aria-describedby="basic-addon3">
</div>

<div class="input-group mb-3">
  <span class="input-group-text">$</span>
  <input type="text" class="form-control" aria-label="Amount (to the nearest dollar)">
  <span class="input-group-text">.00</span>
</div>
```

Your vanity URL
 

\$

.00

## Button addons

```
<div class="demo">
  <div class="input-group mb-3">
    <button class="btn btn-outline-secondary" type="button" id="button-addon1">Button</button>
    <input type="text" class="form-control" placeholder="" aria-label="Example text with button
      addon" aria-describedby="button-addon1">
    </div>

  <div class="input-group mb-3">
    <input type="text" class="form-control" placeholder="Recipient's username"
      aria-label="Recipient's username" aria-describedby="button-addon2">
    <button class="btn btn-outline-secondary" type="button" id="button-addon2">Button</button>
    </div>

  <div class="input-group mb-3">
    <button class="btn btn-outline-secondary" type="button">Button</button>
    <button class="btn btn-outline-secondary" type="button">Button</button>
    <input type="text" class="form-control" placeholder="" aria-label="Example text with two
      button addons">
    </div>

  <div class="input-group">
    <input type="text" class="form-control" placeholder="Recipient's username"
      aria-label="Recipient's username with two button addons">
    <button class="btn btn-outline-secondary" type="button">Button</button>
    <button class="btn btn-outline-secondary" type="button">Button</button>
    </div>
</div>
```

Button

Button

Button

Button

Button

Button



## 7. 表單驗證

is-valid 呼叫 valid-feedback

is-invalid 呼叫 invalid-feedback

required 表示需要驗證

此驗證功能可以搭配後端並該改.is-valid or .is-invalid  
顯示相對的警告訊息！

```
<div class="demo">
  <form class="row g-3">
    <div class="col-md-4">
      <label for="validationServer01" class="form-label">First name</label>
      <input type="text" class="form-control is-invalid" id="validationServer01" value="Mark"
      required>
      <div class="valid-feedback"> 輸入正確!</div>
      <div class="invalid-feedback">錯誤輸入錯誤!</div>
    </div>
    <div class="col-md-4">
      <label for="validationServer02" class="form-label">Last name</label>
      <input type="text" class="form-control is-valid" id="validationServer02" value="Otto"
      required>
      <div class="valid-feedback"> 輸入正確!</div>
      <div class="invalid-feedback">錯誤輸入錯誤!</div>
    </div>
    <div class="col-12">
      <div class="form-check">
        <input class="form-check-input is-invalid" type="checkbox" value="" id="invalidCheck3"
        aria-describedby="invalidCheck3Feedback" required>
        <label class="form-check-label" for="invalidCheck3">
          同意條款和條件
        </label>
        <div id="invalidCheck3Feedback" class="invalid-feedback">你必須先同意條款和條件</div>
      </div>
    </div>
    <div class="col-12">
      <button class="btn btn-primary" type="submit">Submit form</button>
    </div>
  </form>
</div>
```

The screenshot shows a Bootstrap form with two text input fields. The first field, labeled 'First name', contains the text 'Mark' and has a red border with a red error icon and the message '錯誤輸入錯誤!' (Error input error!). The second field, labeled 'Last name', contains the text 'Otto' and has a green border with a green checkmark icon and the message '輸入正確!' (Input correct!). Below these fields is a checkbox labeled '同意條款和條件' (I agree with the terms and conditions) with the message '你必須先同意條款和條件' (You must first agree with the terms and conditions). At the bottom is a blue 'Submit form' button.

## 8. Floating labels(浮動標籤)

```
<div class="demo">
  <div class="form-floating mb-3">
    <input type="email" class="form-control" id="floatingInput" placeholder="name@example.com">
    <label for="floatingInput">Email address</label>
  </div>
  <div class="form-floating">
    <input type="password" class="form-control" id="floatingPassword" placeholder="Password">
    <label for="floatingPassword">Password</label>
  </div>
</div>
```

The screenshot shows a Bootstrap form with two input fields. The first field is labeled 'Email address' and has a blue border. The second field is labeled 'Password' and has a white border. The labels are positioned above the input fields and are not floating.

## 9. 表單排版練習

### 利用格線排版, 搭配gutter

```
<div class="demo">
  <div class="row mb-3">
    <div class="col-6">
      <input type="text" class="form-control" placeholder="First name">
    </div>
    <div class="col-6">
      <input type="text" class="form-control" placeholder="Last name">
    </div>
  </div>
  <div class="row gx-1">
    <div class="col-6">
      <input type="text" class="form-control" placeholder="First name">
    </div>
    <div class="col-6">
      <input type="text" class="form-control" placeholder="Last name">
    </div>
  </div>
</div>
```

First name Last name

First name Last name

```
<div class="demo">
  <div class="row gx-2 gy-3">
    <div class="col-6">
      <label for="" class="form-text">First name</label>
      <input type="text" class="form-control" placeholder="First name">
    </div>
    <div class="col-6">
      <label for="" class="form-text">Last name</label>
      <input type="text" class="form-control" placeholder="Last name">
    </div>
    <div class="col-12">
      <label for="" class="form-text">email</label>
      <input type="email" class="form-control" placeholder="email">
    </div>
    <div class="col-12">
      <label for="" class="form-text">address</label>
      <input type="text" class="form-control" placeholder="address">
    </div>
  </div>
</div>
```

First name Last name

First name Last name

email

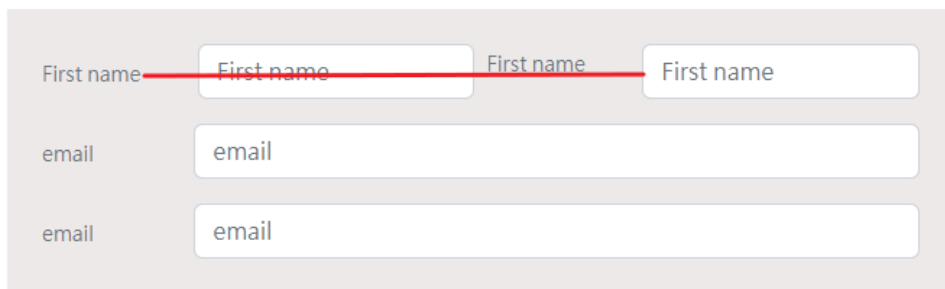
email

address

address

## label文字垂直置中 col-form-label

```
<div class="demo">
  <div class="row gx-2 gy-3">
    <div class="col-6">
      <div class="row">
        <label for="" class="form-text col-sm-4 col-form-label">First name</label>
        <div class="col-sm-8">
          <input type="text" class="form-control" placeholder="First name">
        </div>
      </div>
    </div>
    <div class="col-6">
      <div class="row">
        <label for="" class="form-text col-sm-4">First name</label>
        <div class="col-sm-8">
          <input type="text" class="form-control" placeholder="First name">
        </div>
      </div>
    </div>
    <div class="col-12">
      <div class="row">
        <label for="" class="form-text col-sm-2 col-form-label">email</label>
        <div class="col-sm-10">
          <input type="email" class="form-control" placeholder="email">
        </div>
      </div>
    </div>
  </div>
</div>
```



## 11. 使用後端練習(json-server)

<https://www.npmjs.com/package/json-server>

### 檢查是否已安裝 Node.js

```
C:\Users\Richie\Desktop\json-server-demo>npm -v
10.7.0
```

### 安裝 json-server

```
C:\Users\Richie\Desktop\json-server-demo>npm install -g json-server
changed 45 packages in 873ms
14 packages are looking for funding
  run 'npm fund' for details
C:\Users\Richie\Desktop\json-server-demo>json-server --version
1.0.0-beta.3
C:\Users\Richie\Desktop\json-server-demo>
```

### 建立假資料庫：db.json

準備一個 JSON 檔，當作「假資料庫」。

```

() dbjson
() dbjson > [ ] products > {} 1
1 {
2   "products": [
3     {
4       "id": "1",
5       "name": "智慧手錶",
6       "price": 6000
7     },
8     {
9       "price": 6900,
10      "id": "2"
11    }
12  ],
13  "users": [
14    {
15      "id": "1",
16      "username": "test",
17      "password": "1234"
18    }
19  ]
20 }

```

## 啟動 json-server

```

C:\Users\Richie\Desktop\json-server-demo>json-server --watch db.json --port 3000
--watch/-w can be omitted, JSON Server 1+ watches for file changes by default
JSON Server started on PORT :3000
Press CTRL-C to stop
Watching db.json...

🐉 🍷 🍷

Index:
http://localhost:3000/

Static files:
Serving ./public directory if it exists

Endpoints:
http://localhost:3000/products
http://localhost:3000/users

```

```

← → ↺ 🏠 ⓘ localhost:3000

```

### JSON Server

🍷. 9('🍷') 🍷.

/products - 4 items

/users - 1 item

## json-server 自動產生的 CRUD 路由

| 動作   | HTTP 方法 | 路徑          | 說明            |
|------|---------|-------------|---------------|
| C 新增 | POST    | /products   | 新增一筆產品        |
| R 全部 | GET     | /products   | 取得所有產品        |
| R 單筆 | GET     | /products/1 | 取得 id=1 的產品   |
| U 全部 | PUT     | /products/1 | 完整更新 id=1 的資料 |
| U 局部 | PATCH   | /products/1 | 局部更新 id=1 的資料 |
| D 刪除 | DELETE  | /products/1 | 刪除 id=1 的產品   |